

Comments on

Fully accounted NLO electromagnetic radiative corrections to the width of J/ψ decay

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Equation and table numbers refer to the current draft (`w_jpsi-4`). “VPK note” refers to *Radiative Corrections to the J/ψ photoproduction reaction*.

1 Eq. (1) and Eq. (17): replace E_V by m_V

Both read $1/(2E_V)$. The whole calculation is done in the J/ψ rest frame, where $E_V = m_V$ identically, and the text already says so. Writing m_V removes a symbol that never takes any other value, and makes the normalisation of Eq. (1) directly comparable with the standard two-body form.

2 Eq. (6): use the factorised form, and add the ultrarelativistic limit

Eq. (6) is correct — it is algebraically identical to Eq. (7) of the VPK note (verified symbolically; the difference is exactly zero). But the factorised form is easier to read and to check dimensionally, because the mass dependence enters only through dimensionless ratios:

$$\Gamma_B = \frac{4\pi\alpha^2}{f_V^2} \frac{m_V}{3} \left(1 + \frac{2m_\ell^2}{m_V^2}\right) \sqrt{1 - \frac{4m_\ell^2}{m_V^2}}.$$

Please also add the ultrarelativistic limit, Eq. (8) of the VPK note, which is the form actually used to fix f_V from the measured leptonic width:

$$\Gamma_B = \frac{4\pi\alpha^2}{f_V^2} \frac{m_V}{3}.$$

Worth stating explicitly: the correction connecting the two is *second* order in the mass ratio, not first. With $x = m_\ell^2/m_V^2$,

$$(1 + 2x)\sqrt{1 - 4x} = 1 - 6x^2 + O(x^3),$$

so the linear term cancels. This is why the massless limit is so accurate: 4×10^{-15} for $J/\psi \rightarrow e^+e^-$ and 8×10^{-6} for $J/\psi \rightarrow \mu^+\mu^-$. It reaches only 2×10^{-3} for $\rho \rightarrow \mu^+\mu^-$.

3 Table I: please say what $m_V = 1.5$ GeV is

Tables I and II are computed at $m_V = 1.5$ GeV, which is neither the J/ψ nor any physical vector meson. A reader meeting it just after Section II, which sets $m_V = 3.0969$ GeV, will be confused.

We understand it to be the last row of Table I of Ref. [10], whose tabulated column is $2E$ (MeV), running from 500 to 1500 with ρ , ω and ϕ marked — that is, $2E$ is the meson mass, since $E = m_V/2$ at resonance (stated in [10] below their Eq. (22)). One sentence would settle it: “for comparison with Ref. [10] we take $m_V = 1.5$ GeV, the largest mass tabulated there.”

4 The comparison with Ref. [10] does not hold where it is claimed

The text states that Table I can be “successfully compared” with Table I of Ref. [10] at $\omega/m_e = 0.01$, last line of the first half. At that line the two disagree, and the disagreement grows with r_{cut} :

r_{max}	Ref. [10]	this work	difference
0.10	−0.087	−0.0866	~ 0
0.25	−0.034	−0.0330	0.001
0.50	−0.002	+0.000044	0.002
1.00	+0.021	+0.023891	0.0029

The soft part agrees exactly (both give $\delta_R = -0.747$), so the mass and the cut-off are matched correctly. The discrepancy is entirely in the hard part and equals $5\alpha/4\pi = 0.0029$ at $r_{max} = 1$.

The reason is that **Eq. (24) of Ref. [10] is missing a term**. Their differential spectrum, Eq. (23), is correct and agrees with Eq. (14) of this work; the term was lost when that spectrum was integrated analytically. Their Eq. (26), and their Table I, inherit the omission. Recomputing their table from their own Eq. (26) reproduces every printed digit, electrons and muons alike, which confirms the diagnosis.

Suggested replacement for the claim of agreement:

Our Eq. (14) agrees with expression (23) of [10]. The integrated form, Eq. (24) of [10], is however missing a term $(\alpha/\pi)(3r_{max}/2 - r_{max}^2/4)$, so that the totals tabulated in Table I of [10] are low by that amount; at $r_{max} = 1$ the deficit is $5\alpha/4\pi = 0.0029$. Our results agree with [10] in the small- r_{max} rows, where the missing term lies below the quoted precision.

5 Please add the closed-form total correction

The paper gives δ_{self} (8), δ_{vert} (12), δ_{soft} (13), and δ_{hard} only as an unevaluated integral (14), then the exact result (28). There is no compact closed form for the total ultrarelativistic correction, which is what an experiment actually applies. We would like it in the paper. The correct expression — Eq. (24) of Ref. [10] with the missing term restored — is

$$\delta(m_V, \omega_{cut}) = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{m_V}{m_\ell} \right) \left(\ln r + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r) - \frac{\pi^2}{6} \right) + \frac{17}{72} + R(r) + \frac{r^2}{16} - \frac{3}{8}r \right]$$

with

$$r = \frac{2\omega_{cut}}{m_V}, \quad R(r) = \left(r - \frac{r^2}{4} - \frac{3}{4}\right) \left[\ln\left(\frac{m_V}{m_\ell} \sqrt{1-r}\right) - \frac{1}{2} \right], \quad L_2(x) = - \int_0^x \frac{\ln(1-t)}{t} dt.$$

For the muon channel add the electron-loop vacuum polarization, which is exactly δ_{self} of Eq. (8) evaluated with m_e .

The last two terms, $r^2/16 - \frac{3}{8}r$, are the ones absent from [10]. Checks this expression passes:

- it reproduces Table I of this work to all six digits at every r_{cut} , in both channels, and is independent of ω by construction;
- at $r = 1$ it gives exactly $\delta_{self} + 3\alpha/4\pi$, i.e. δ_{self} plus the FSIC of Eq. (16);
- for $J/\psi \rightarrow e^+e^-$ at $r = 1$ it gives 0.026135, matching the Bardin–Shumeiko result of Table III;
- for $J/\psi \rightarrow \mu^+\mu^-$ at $r = 1$ it gives 0.034017 against the exact 0.034097 — the 8×10^{-5} residue is the ultrarelativistic approximation itself, consistent with the accuracy discussion of Section IV.

6 Reference [9]

[9] P. Chatagnon, e-Print: 2602.22128 [hep-ex] (2026).

is now published. Please replace with

[9] P. Chatagnon, Phys. Rev. C 113 (2026) 6, 065203.

Additional point, for consideration

Eq. (4). As printed,

$$\sum_{\text{pol}} \varepsilon_\alpha(p_V) \varepsilon_\beta(p_V) = -g_{\alpha\beta}$$

is the polarization sum for a *massless* vector. For a massive vector meson it is

$$\sum_\lambda \varepsilon_\alpha \varepsilon_\beta^* = -g_{\alpha\beta} + \frac{p_\alpha p_\beta}{m_V^2}.$$

The extra term does drop here, because it contracts the lepton current with $p_V = p_1 + p_2$ and $\bar{u}(p_1)(\not{p}_1 + \not{p}_2)v(p_2) = 0$ by the Dirac equations — so Eq. (5) and everything after it are correct. But the identity as written is not true on its own, and the surrounding text calls it “the photon” while the sum is over the three J/ψ polarizations. A clause such as “the $p_\alpha p_\beta$ term is dropped by current conservation” would fix it.